برنامج موهبة الإثرائي البحثي
جامعة الملك فهد للبترول والمعادن
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NEXT-GENERATION DESALINATION: 4TH REFRIGERANTS GENERATION AND HYDRATE TECHNOLOGY FOR SUSTAINABLE FRESH WATER

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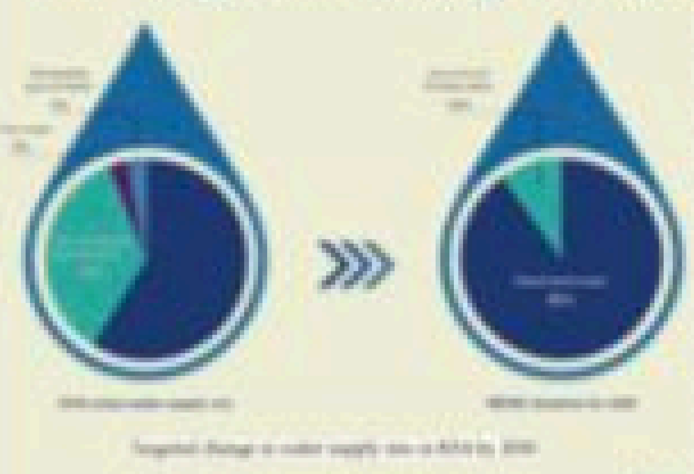
MOTIVATION

- Approximately **one billion people** face water scarcity or inadequate water infrastructure.
- Seawater **desalination**, crucial in regions like the Middle East including **KSA**, is vital for freshwater access and is expected to grow in importance.(1)
- Innovation in desalination technologies is crucial for meeting increasing demand.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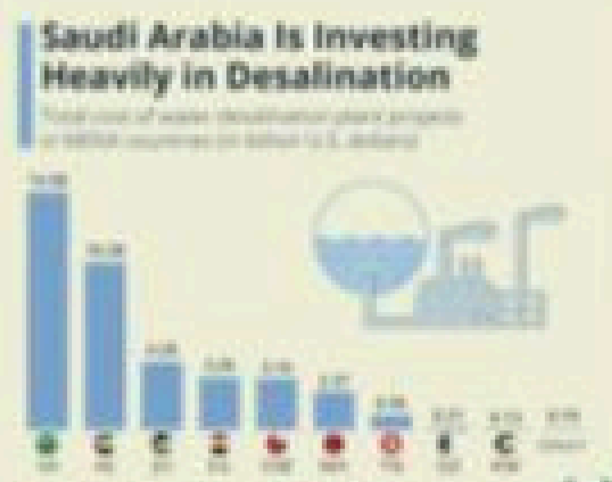

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BACKGROUND

HYDRATES

- Crystalline solids form when water molecules encase a guest molecule, creating a frozen cage. They form under low temperatures and high pressure.

REFRIGERANTS

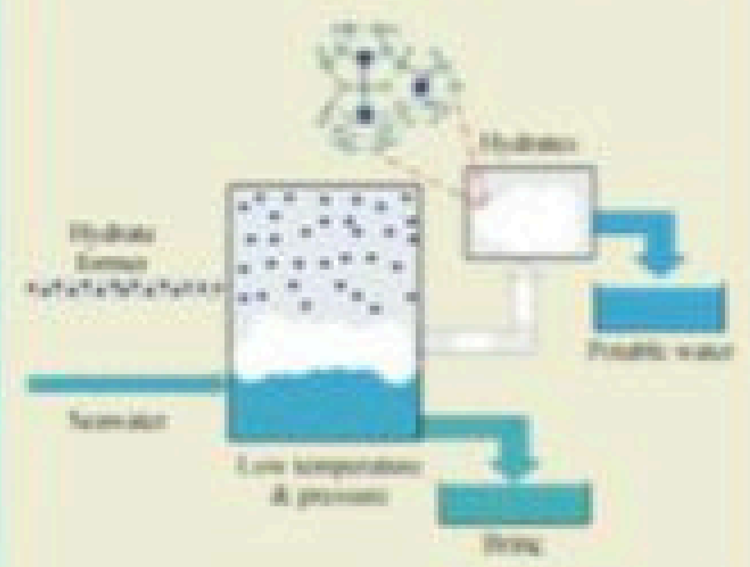
- Substances used in cooling systems to transfer heat (e.g. HFCs). They are also used as guest molecules

DESALINATION

- Process that removes salts from seawater.
- Common methods:
  1. Thermal
  2. Membrane-based

Hydrate-based desalination

- Recently, there has been a growing interest in hydrate-based desalination (HBD) technologies, which recover water in the form of gas or hydrate crystals.



OBJECTIVE

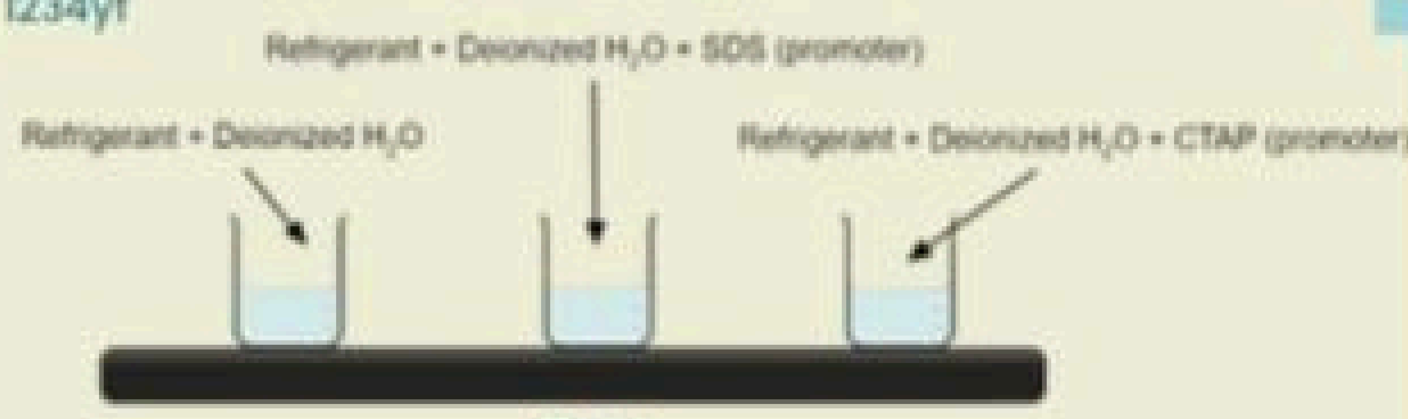
- To identify and select refrigerants suitable for hydrate formation to be used in desalination.
- To assess the efficiency and stability of hydrates formed using these refrigerants.
- To investigate the role of different promoters in enhancing hydrate formation.

METHODOLOGY

Materials

Refrigerants selected:
- 134a
- 410a
- 1234yf

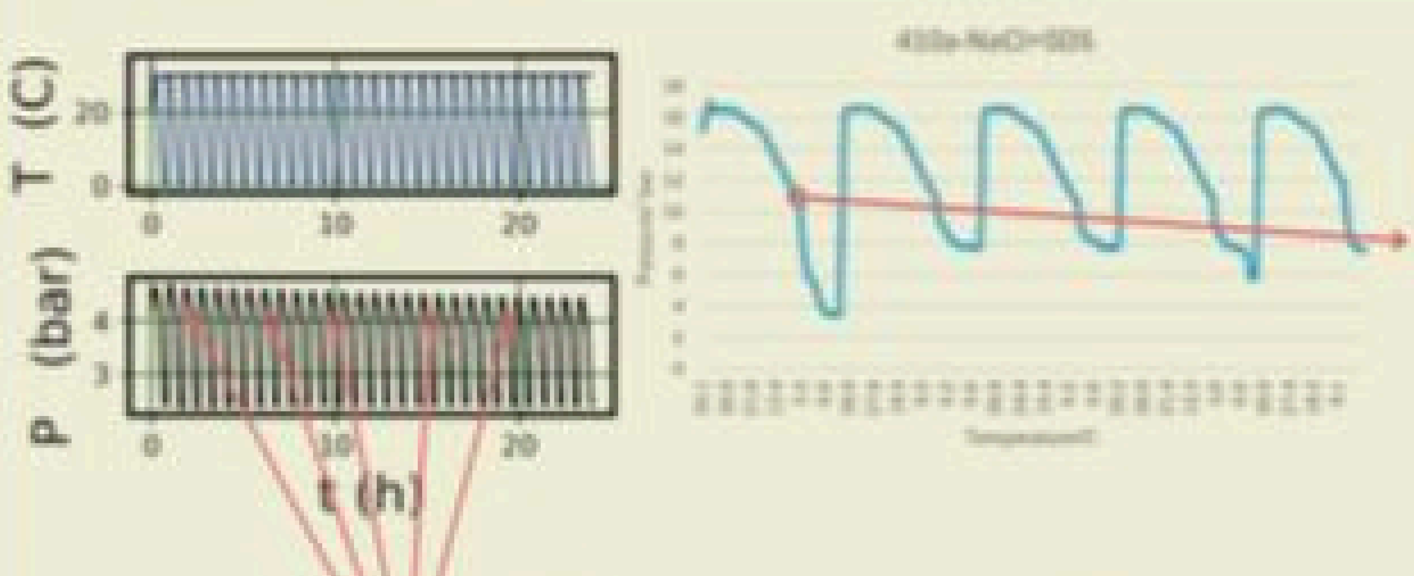
Promoters selected:
- Sodium dodecyl sulfate (SDS)
- Cetyltrimethylammonium bromide (CTAB)



System used:
High pressure stirred automated lag time apparatus (HPS-ALTA)

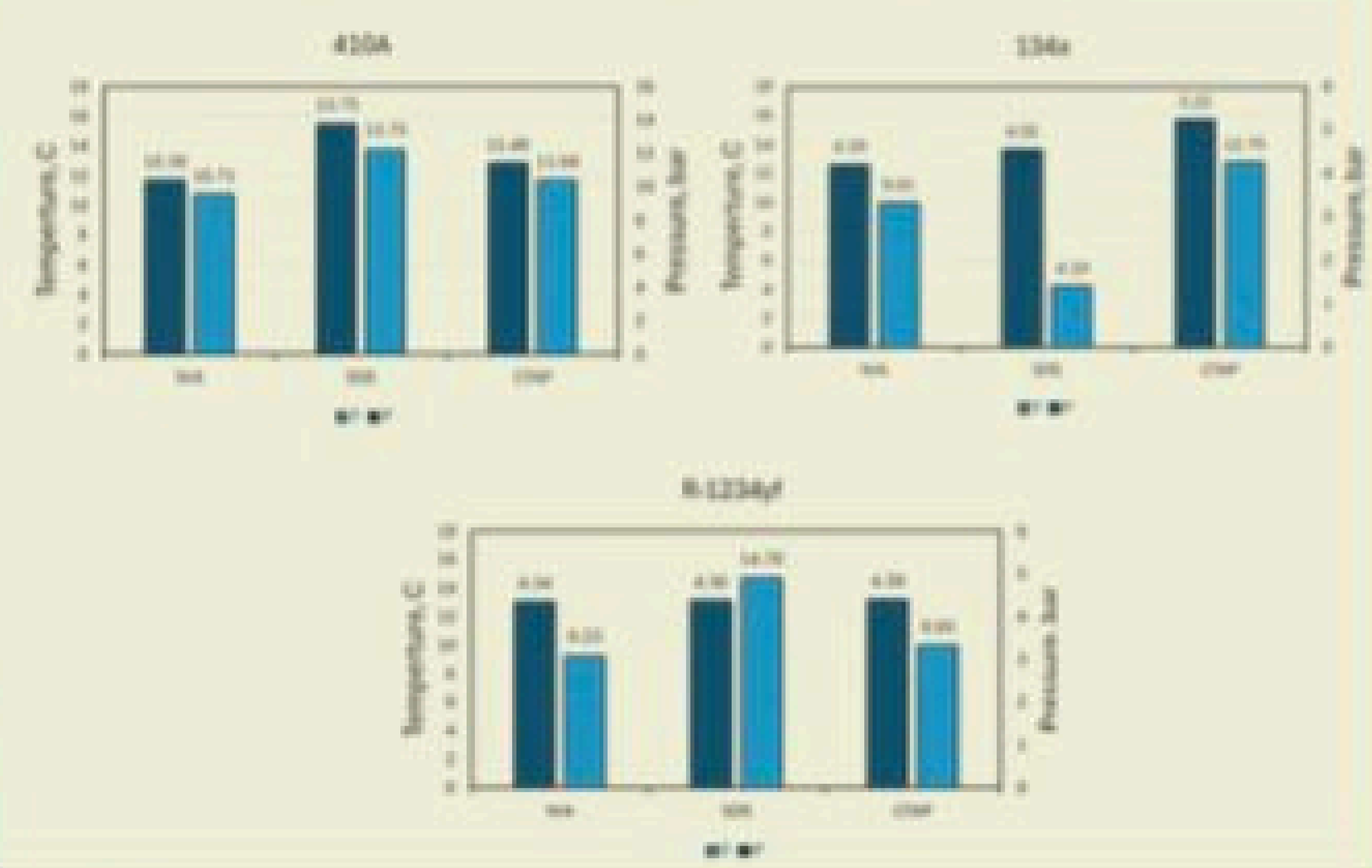
- Simultaneous Experiments
- Temperature and Pressure Control
- Temperature and pressure Sensors
- Rapid Heating and Cooling

ANALYSIS



- Step 1: 5 random cycles with pressure drop are taken from the 24 cycles and the pressure drop point in each cycle is found.
- Step 2: A pressure/temperature graph is made to determine the hydrate onset formation point conditions.
- Step 3: The point of hydrate formation is plotted, and results can be compared to check the efficiency of materials.

RESULTS AND DISCUSSION



CONCLUSION

- Hydrate-based desalination has the **highest potential** to be the future solution for **sustainable freshwater production** globally, and specifically for Saudi Arabia.
- The fourth-generation refrigerant (1234yf), with a GWP of less than 1, has demonstrated superior efficiency and high potential for use in hydrate-based desalination.
- The findings underscore the environmental and economic advantages of these eco-friendly refrigerants and promoters, highlighting their effectiveness.

FUTURE WORK

1. Sensitivity analysis on different parameters: salt type and concentration, other equipment, promoters, refrigerants.
2. Modeling: thermodynamic and molecular.

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